A PROJECT TITLE

ON

Crop Yield Prediction Using Ridge, Lasso, and Elastic Net Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Crop Yield Prediction Using Ridge, Lasso, and Elastic Net Regression

1. Abstract:

Agricultural productivity depends on a complex interplay of climatic, soil, and management factors, making crop yield forecasting a persistent challenge for farmers, agribusinesses, and policymakers. This project develops and compares three regularized linear regression techniques, namely Ridge Regression, Lasso Regression, and Elastic Net Regression, for predicting crop yield from historical agricultural data. The dataset used contains features such as rainfall, average temperature, humidity, soil nutrient levels including nitrogen, phosphorus, and potassium, and pesticide and fertilizer usage across different crop types and growing seasons. Since such datasets typically contain many correlated predictor variables, ordinary least squares regression tends to overfit and produce unstable coefficient estimates. Ridge Regression addresses this by applying an L2 penalty that shrinks coefficients and reduces variance while retaining every feature. Lasso Regression instead applies an L1 penalty that can shrink some coefficients exactly to zero, performing automatic feature selection and yielding a more interpretable model. Elastic Net Regression combines both penalties, balancing the stability of Ridge with the sparsity of Lasso, and performs particularly well when predictors such as weather and soil variables are highly correlated. The methodology involves data cleaning, treatment of missing values, encoding of categorical variables, feature scaling, and splitting the dataset into training and testing subsets. Each model is trained on the training subset, with its regularization strength tuned through cross-validation to minimize prediction error on unseen data. Performance is then evaluated and compared using metrics such as R-squared, Mean Absolute Error, and Root Mean Squared Error. The comparative analysis highlights the trade-off between bias and variance across the three strategies and identifies which approach yields the most accurate and generalizable predictions for the dataset under study. The resulting model helps stakeholders anticipate yield outcomes well in advance, supporting more informed decisions around irrigation scheduling, fertilizer allocation, and market supply planning, while demonstrating the practical value of regularized regression methods in real-world agricultural decision-making.

2. INTRODUCTION:

Agriculture continues to be the backbone of many economies, and accurately forecasting crop yield is essential for effective planning around food security, resource allocation, and market pricing. Yield outcomes are influenced by a wide range of interdependent factors, including rainfall, temperature, soil nutrient content, and farming practices, which makes the relationship difficult to capture using simple linear assumptions. Traditional approaches to yield estimation have often relied on historical averages and expert judgement, which can fail to account for the non-linear and correlated nature of agricultural data, especially as climate variability increases.

Machine learning offers a data-driven alternative that can learn patterns directly from historical records rather than relying on fixed assumptions. Among the available techniques, regularized linear regression models such as Ridge, Lasso, and Elastic Net are particularly well suited to agricultural datasets, where predictor variables such as weather parameters and soil nutrients are often highly correlated with one another. Regularization helps control model complexity and reduces overfitting, and in the case of Lasso and Elastic Net, can also identify the most influential features driving yield outcomes. This project applies and compares these three regression techniques to build a reliable crop yield prediction model and to study how each one responds to multicollinearity and varying levels of regularization strength.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.8 or above
- Development Environment: Jupyter Notebook, Google Colab, or Visual Studio Code
- Core Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, and Seaborn
- Version Control: Git and GitHub for source code management
- Documentation Tool: Microsoft Word for report preparation
- Hardware: A standard laptop or desktop with at least 4 GB RAM (8 GB recommended for faster model training)

Signature of the Guide Course Instructor